

Quantum gases and condensates

30

Exchange symmetry affects the occupation of allowed states in quantum gases. If the density of the gas is very low, such that $n\lambda_{\text{th}}^3 \ll 1$, we can ignore this and forget about exchange symmetry; this is what we do for gases at room temperature. But if the density is high, the effects of exchange symmetry become very important and it really starts to matter whether the particles you are considering are fermions or bosons. In this chapter, we consider quantum gases in detail and explore the possible effects that one can observe.

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30.1 The non-interacting quantum fluid

We first consider a fluid composed of non-interacting particles. To keep things completely general for the moment, we will consider particles with spin S . This means that each allowed momentum state is associated with $2S + 1$ possible spin states.¹ If we can ignore interactions between particles, the grand partition function $\mathcal{Z}$ is simply the product of single-particle partition functions, so that

$$\mathcal{Z} = \prod_{\mathbf{k}} \mathcal{Z}_{\mathbf{k}}^{2S+1}, \quad (30.1)$$

where

$$\mathcal{Z}_{\mathbf{k}} = (1 \pm e^{-\beta(E_{\mathbf{k}} - \mu)})^{\pm 1} \quad (30.2)$$

is a single particle partition function and where the $\pm$ sign is $+$ for fermions and $-$ for bosons.²

¹If the spin is S , there are $2S + 1$ possible states corresponding to the z -component of angular momentum being $-S, -S + 1, \dots, S$.

²These results follow directly from eqns 29.18 and 29.20.

Example 30.1

Find the grand potential for a three-dimensional gas of non-interacting bosons and fermions with spin S .

Solution:

The grand potential Φ_G is obtained from eqn 30.1 as follows:

$$\begin{aligned} \Phi_G &= -k_B T \ln \mathcal{Z} \\ &= \mp k_B T (2S + 1) \sum_{\mathbf{k}} \ln(1 \pm e^{-\beta(E_{\mathbf{k}} - \mu)}) \\ &= \mp k_B T (2S + 1) \int_0^\infty \ln(1 \pm e^{-\beta(E - \mu)}) g(E) dE, \quad (30.3) \end{aligned}$$

where $g(E)$ is the density of states, which can be derived as follows. States in k -space are uniformly distributed, and so

$$g(k) dk = \frac{4\pi k^2 dk}{(2\pi/L)^3} \times (2S+1) = \frac{(2S+1)V k^2 dk}{2\pi^2}, \quad (30.4)$$

where $(2S+1)$ is the spin degeneracy factor and $V = L^3$ is the volume. Using $E = \hbar^2 k^2/2m$ we can transform this into

$$g(E) dE = \frac{(2S+1)V E^{1/2} dE}{(2\pi)^2} \left(\frac{2m}{\hbar^2}\right)^{3/2}, \quad (30.5)$$

and hence

$$\Phi_G = \mp k_B T \frac{(2S+1)V}{(2\pi)^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^\infty \ln(1 \pm e^{-\beta(E-\mu)}) E^{1/2} dE, \quad (30.6)$$

which after integrating by parts yields

$$\Phi_G = -\frac{2}{3} \frac{(2S+1)V}{(2\pi)^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^\infty \frac{E^{3/2} dE}{e^{-\beta(E-\mu)} \pm 1}. \quad (30.7)$$

³Note that in the derived expressions, the $\pm$ sign means $+$ for fermions and $-$ for bosons.

The grand potential evaluated in the previous example can be used to derive various thermodynamic functions for fermions and bosons.³ Another way to get to the same result is to evaluate the mean occupation $n_{\mathbf{k}}$ of a state with wave vector $\mathbf{k}$, which is given by

$$n_{\mathbf{k}} = k_B T \frac{\partial}{\partial \mu} Z_{\mathbf{k}} = \frac{1}{e^{\beta(E_{\mathbf{k}}-\mu)} \pm 1}, \quad (30.8)$$

and then use this expression to derive directly quantities such as

$$N = \sum_{\mathbf{k}} n_{\mathbf{k}} = \int_0^\infty \frac{g(E) dE}{e^{\beta(E-\mu)} \pm 1}, \quad (30.9)$$

and

$$U = \sum_{\mathbf{k}} n_{\mathbf{k}} E_{\mathbf{k}} = \int_0^\infty \frac{E g(E) dE}{e^{\beta(E-\mu)} \pm 1}. \quad (30.10)$$

For reasons which will become more clear below, we will write $e^{\beta\mu}$ as the **fugacity** z , i.e.

$$z = e^{\beta\mu}. \quad (30.11)$$

These give expressions for N and U as follows:

$$N = \left[\frac{(2S+1)V}{(2\pi)^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \right] \int_0^\infty \frac{E^{1/2} dE}{z^{-1} e^{\beta E} \pm 1} \quad (30.12)$$

and

$$U = \left[\frac{(2S+1)V}{(2\pi)^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \right] \int_0^\infty \frac{E^{3/2} dE}{z^{-1} e^{\beta E} \pm 1}. \quad (30.13)$$

One problem with all these types of formula, such as eqns 30.7, 30.12 and 30.13, is that to simplify them any further, you have to do a difficult integral. Fortunately, we can show that these integrals are related to the **polylogarithm** function $\text{Li}_n(x)$ (see Appendix C.5), so that

$$\int_0^\infty \frac{E^{n-1} dE}{z^{-1} e^{\beta E} \pm 1} = (k_B T)^n \Gamma(n) [\mp \text{Li}_n(\mp z)], \quad (30.14)$$

where $\Gamma(n)$ is a gamma function. This result is proved in the appendix (eqn C.36). The crucial thing to realize is that $\text{Li}_n(z)$ is just a numerical function of z , i.e. of the temperature and the chemical potential. This integral then allows us to establish, after a small amount of algebra, that the number N of particles is given by

$$N = \frac{(2S+1)V}{\lambda_{\text{th}}^3} [\mp \text{Li}_{3/2}(\mp z)], \quad (30.15)$$

and the internal energy U is given by

$$\begin{aligned} U &= \frac{3}{2} k_B T \frac{(2S+1)V}{\lambda_{\text{th}}^3} [\mp \text{Li}_{5/2}(\mp z)] \\ &= \frac{3}{2} N k_B T \frac{\text{Li}_{5/2}(\mp z)}{\text{Li}_{3/2}(\mp z)}. \end{aligned} \quad (30.16)$$

We will use these equations in subsequent sections. Note also that we have from eqns 30.7 and 30.13 that

$$\Phi_G = -\frac{2}{3} U. \quad (30.17)$$

Example 30.2

Evaluate N , U and Φ_G (from eqns 30.15, 30.16 and 30.17) in the high-temperature limit.

Solution:

In the high-temperature limit, namely $\beta\mu \ll 1$, we can use the fact that $\text{Li}_n(z) \approx z$ when $|z| \ll 1$. Hence

$$N \approx \frac{(2S+1)V}{\lambda_{\text{th}}^3}, \quad (30.18)$$

$$U \approx \frac{3}{2} N k_B T, \quad (30.19)$$

$$\Phi_G \approx -N k_B T. \quad (30.20)$$

These three equations are reassuringly familiar. The equation for N shows that the number density of particles N/V is such that, on average, $2S+1$ particles (one for each spin state) occupy a volume λ_{th}^3 . The equation for U asserts that the energy per particle is the familiar equipartition result $\frac{3}{2} k_B T$. The equation for Φ_G , together with $\Phi_G = -pV$ (from eqn 22.49) yields the ideal gas law $pV = N k_B T$.

30.2 The Fermi gas

What we have done so far is to consider bosons and fermions on an equal footing. Let us now restrict our attention to a gas of fermions (known as a **Fermi gas**) and to get a feel for what is going on, let us also consider $T = 0$. Fermions will occupy the lowest-energy states, but we can only put one fermion in each state, and thus only $2S + 1$ in each energy level. The fermions will fill up the energy levels until they get to an energy E_F , known as the **Fermi energy**, which is the energy of the highest occupied state at a temperature of absolute zero.⁴ Thus we define

$$E_F = \mu(T = 0). \quad (30.21)$$

This makes sense because $\mu(T = 0) = \partial E / \partial N$, which gives $\mu(T = 0) = E(N) - E(N - 1) = E_F$. At absolute zero, we have that $\beta \rightarrow \infty$, and hence the occupation $n_{\mathbf{k}}$ is given by

$$n_{\mathbf{k}} = \frac{1}{e^{\beta(E_{\mathbf{k}} - \mu)} + 1} = \theta(\mu - E_{\mathbf{k}}), \quad (30.22)$$

where $\theta(x)$ is a Heaviside step function.⁵ At absolute zero, therefore, the number of states is given by

$$N = \int_0^{k_F} g(\mathbf{k}) d^3k, \quad (30.23)$$

where k_F is the **Fermi wave vector**, defined by

$$E_F = \frac{\hbar^2 k_F^2}{2m}. \quad (30.24)$$

Hence the number of fermions N is given by

$$N = \frac{(2S + 1)V}{2\pi^2} \frac{k_F^3}{3}, \quad (30.25)$$

so that writing $n = N/V$, we have

$$k_F = \left[\frac{6\pi^2 n}{2S + 1} \right]^{1/3}, \quad (30.26)$$

and hence

$$E_F = \frac{\hbar^2}{2m} \left[\frac{6\pi^2 n}{2S + 1} \right]^{2/3}. \quad (30.27)$$

Example 30.3

Evaluate k_F and E_F for spin- $\frac{1}{2}$ particles.

Solution:

When $S = \frac{1}{2}$, $2S + 1 = 2$ and hence eqns 30.26 and 30.27 become

$$k_F = [3\pi^2 n]^{1/3}, \quad (30.28)$$

and

$$E_F = \frac{\hbar^2}{2m} [3\pi^2 n]^{2/3}. \quad (30.29)$$

⁴The highest filled energy level at $T = 0$ is known as the **Fermi level**, though this can be a misleading term as, for example in semiconductors, there may not be any states at the chemical potential (which lies somewhere in the energy gap).

⁵The Heaviside step function $\theta(x)$ is defined by

$$\theta(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$$

It is plotted in Fig. 30.1.

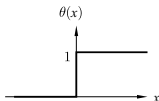


Fig. 30.1 The Heaviside step function.

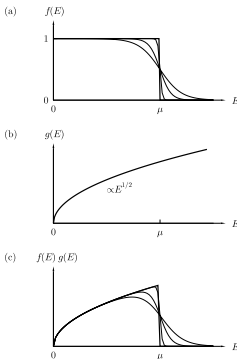


Fig. 30.2 (a) The Fermi function $f(E)$ defined by eqn 29.26. The thick line is for $T = 0$. The step function is smoothed out as the temperature is increased (shown as thinner lines). The temperatures shown are $T = 0$, $T = 0.01\mu/k_B$, $T = 0.05\mu/k_B$ and $T = 0.1\mu/k_B$. (b) The density of states $g(E)$ for a non-interacting fermion gas in three dimensions is proportional to $E^{1/2}$. (c) $f(E)g(E)$ for the same temperatures as in (a).

At $T = 0$, the distribution function $f(E)$ is a Heaviside step function, taking the value 1 for $E < \mu$ and 0 for $E > \mu$. This step is smoothed out as the temperature T increases, as shown in Fig. 30.2(a). The density of states $g(E)$ for a non-interacting fermion gas in three dimensions is proportional to $E^{1/2}$ (as shown in eqn 30.4) and this is plotted in Fig. 30.2(b). The product of $f(E)g(E)$ gives the actual number distribution of fermions, and this is shown in Fig. 30.2(c). The sharp cutoff you would expect at $T = 0$ is smoothed over an energy scale $k_B T$ around the chemical potential μ .

The electrons in a metal can be treated as a non-interacting gas of fermions. Using the number density n of electrons in a metal, one can calculate the Fermi energy using eqn 30.29, and some example results are shown in Table 30.1. The Fermi energies are all several eV; converting each number into a temperature, the so-called **Fermi temperature** $T_F = E_F/k_B$, yields values of several tens of thousands of Kelvin. Thus the Fermi energy is a large energy scale, and hence for most metals the Fermi function is close to a step function, at pretty much all temperatures below their melting temperature. In this case, the electrons in a metal are said to be in the **degenerate limit**.

The pressure of these electrons is given (by using eqns 22.49 and 30.17) as

$$p = \frac{2U}{3V}, \quad (30.30)$$

	n (10^{28} m^{-3})	E_F (eV)	$\frac{2}{3}nE_F$ (10^9 N m^{-2})	B (10^9 N m^{-2})
Li	4.70	4.74	23.8	11.1
Na	2.65	3.24	9.2	6.3
K	1.40	2.12	3.2	3.1
Cu	8.47	7.00	63.3	137.8
Ag	5.86	5.49	34.3	103.6

Table 30.1 Properties of selected metals

as is appropriate for non-relativistic electrons (see Table 25.1). The mean energy of the electrons at $T = 0$ is given by

$$\langle E \rangle = \frac{\int_0^{E_F} E g(E) dE}{\int_0^{E_F} g(E) dE}, \quad (30.31)$$

which with $g(E) \propto E^{1/2}$ gives $\langle E \rangle = \frac{3}{5}E_F$. Writing $U = n\langle E \rangle$, we have that the bulk modulus B is

$$B = -V \frac{\partial p}{\partial V} = \frac{10U}{9V} = \frac{2}{3}nE_F. \quad (30.32)$$

This expression is evaluated in Table 30.1 and gives results which are of the same order of magnitude as experimental values.

The next example computes an integral which is useful for considering analytically the effect of finite temperature.

Example 30.4

Evaluate the integral $I = \int_0^\infty \phi(E)f(E) dE$ as a power series in temperature.

Solution:

Consider the function $\psi(E) = \int_0^E \phi(E') dE'$, which is defined so that $\phi(E) = d\psi/dE$ and therefore

$$\begin{aligned} I = \int_0^\infty \frac{d\psi}{dE} f(E) dE &= [f(E)\psi(E)]_0^\infty - \int_0^\infty \psi(E) \frac{df}{dE} dE \\ &= - \int_0^\infty \psi(E) \frac{df}{dE} dE. \end{aligned} \quad (30.33)$$

Now put $x = (E - \mu)/k_B T$ and hence

$$\frac{df}{dE} = -\frac{1}{k_B T} \frac{e^x}{(e^x + 1)^2}. \quad (30.34)$$

Writing $\psi(E)$ as a power series in x as

$$\psi(E) = \sum_{s=0}^{\infty} \frac{x^s}{s!} \left(\frac{d^s \psi}{dx^s} \right)_{x=0}, \quad (30.35)$$

we can express I as a power series of integrals as follows:

$$I = \sum_{s=0}^{\infty} \frac{1}{s!} \left(\frac{d^s \psi}{dx^s} \right)_{x=0} \int_{-E_F/k_B T}^{\infty} \frac{x^s e^x dx}{(e^x + 1)^2}. \quad (30.36)$$

The integral part of this can be simplified by replacing⁶ the lower limit by $-\infty$. It vanishes for odd s , but for even s

⁶This approximation is valid when $k_B T \gg E_F$.

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{x^s e^x dx}{(e^x + 1)^2} &= 2 \int_0^{\infty} \frac{x^s e^x dx}{(e^x + 1)^2} \\ &= 2 \int_0^{\infty} dx \sum_{n=0}^{\infty} e^x x^s \times [(n+1)(-1)^{n+1} e^{-nx}] \\ &= 2 \sum_{n=1}^{\infty} (-1)^{n+1} n \int_0^{\infty} x^s e^{-nx} dx \\ &= 2(s!) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s} \\ &= 2(s!)(1 - 2^{1-s})\zeta(s), \end{aligned} \quad (30.37)$$

where $\zeta(s)$ is the Riemann zeta function.

Thus the integral is

$$\begin{aligned} I &= \sum_{s=0, s \text{ even}}^{\infty} 2 \left(\frac{d^s \psi}{dx^s} \right)_{x=0} (1 - 2^{1-s})\zeta(s) \\ &= \psi + \frac{\pi^2}{6} \left(\frac{d^2 \psi}{dx^2} \right)_{x=0} + \frac{7\pi^4}{360} \left(\frac{d^4 \psi}{dx^4} \right)_{x=0} + \dots \\ &= \int_{-\infty}^{\mu} \phi(E) dE + \frac{\pi^2}{6} (k_B T)^2 \left(\frac{d\phi}{dE} \right)_{E=\mu} \\ &\quad + \frac{7\pi^4}{360} (k_B T)^4 \left(\frac{d^3 \phi}{dE^3} \right)_{E=\mu} + \dots \end{aligned} \quad (30.38)$$

This expression is known as the **Sommerfeld formula**.

Having derived the Sommerfeld formula, we can now evaluate N and U quite easily. Let us choose $S = \frac{1}{2}$, just to make the equations a little less cumbersome. Then

$$\begin{aligned} N &= \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^{\infty} E^{1/2} f(E) dE \\ &= \frac{V}{3\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \mu^{3/2} \left[1 + \frac{\pi^2}{8} \left(\frac{k_B T}{\mu} \right)^2 + \dots \right], \end{aligned} \quad (30.39)$$

which implies that

$$\mu(T) = \mu(0) \left[1 - \frac{\pi^2}{12} \left(\frac{k_B T}{\mu(0)} \right)^2 + \dots \right]. \quad (30.40)$$

In fact, equating E_F and μ is good to 0.01% for typical metals even at room temperature, although it is worthwhile keeping in the back of one's mind that the two quantities are not the same.

We can also compute the heat capacity of electrons in a metal by a similar technique, as shown in the following example.

Example 30.5

Compute the heat capacity of non-interacting free electrons in a three-dimensional metal.

Solution:

$$\begin{aligned} U &= \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^\infty E^{3/2} f(E) dE \\ &= \frac{V}{5\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \mu(T)^{5/2} \left[1 + \frac{5\pi^2}{8} \left(\frac{k_B T}{\mu(0)} \right)^2 + \dots \right] \\ &= \frac{3}{5} N \mu(T) \left[1 + \frac{\pi^2}{2} \left(\frac{k_B T}{\mu(0)} \right)^2 + \dots \right] \\ &= \frac{3}{5} N \mu(0) \left[1 + \frac{5\pi^2}{12} \left(\frac{k_B T}{\mu(0)} \right)^2 + \dots \right] \end{aligned} \quad (30.41)$$

and hence

$$C_V = \frac{3}{2} N k_B \left(\frac{\pi^2}{3} \frac{k_B T}{\mu(0)} \right) + O(T^3). \quad (30.42)$$

Thus the contribution to the heat capacity from electrons is linear in temperature (recall from Chapter 24 that the heat capacity from lattice vibrations (phonons) is proportional to T^3 at low temperature) and will therefore dominate the heat capacity of a metal at very low temperatures.

⁷The periodic potential which exists in crystalline metals can lead to the formation of energy gaps, i.e. intervals in energy in which there are no allowed states.

The **Fermi surface** is the set of points in k -space whose energy is equal to the chemical potential. If the chemical potential lies in a gap⁷ between energy bands, then the material is a semiconductor or an insulator and there will be no Fermi surface. Thus a metal is a material with a Fermi surface.

30.3 The Bose gas

For the **Bose gas** (a gas composed of bosons), we can use our expressions for N and U in eqns 30.15 and 30.16 to give

$$N = \frac{(2S+1)V}{\lambda_{\text{th}}^3} \text{Li}_{3/2}(z) \quad (30.43)$$

and

$$U = \frac{3}{2} N k_B T \frac{\text{Li}_{5/2}(z)}{\text{Li}_{3/2}(z)}. \quad (30.44)$$

Example 30.6

Evaluate eqns 30.43 and 30.44 for the case $\mu = 0$.

Solution:

If $\mu = 0$ then $z = 1$. Now $\text{Li}_n(1) = \zeta(n)$ where $\zeta(n)$ is the Riemann zeta function. Therefore

$$N = \frac{(2S+1)V}{\lambda_{\text{th}}^3} \zeta\left(\frac{3}{2}\right) \quad (30.45)$$

and

$$U = \frac{3}{2} N k_B T \frac{\zeta(\frac{5}{2})}{\zeta(\frac{3}{2})}. \quad (30.46)$$

The numerical values are $\zeta(\frac{3}{2}) = 2.612$, $\zeta(\frac{5}{2}) = 1.341$, and hence we have that $\zeta(\frac{5}{2})/\zeta(\frac{3}{2}) = 0.513$.

Note that these results will not apply to photons because we have assumed at the beginning that $E = \hbar^2 k^2 / 2m$, whereas for a photon $E = \hbar kc$. This is worked through in the following example.

Example 30.7

Rederive the equation for U for a gas of photons using the formalism of this chapter.

Solution:

The density of states is $g(k) dk = (2S+1)V k^2 dk / (2\pi^2)$. A photon has a spin of 1, but the 0 state is not allowed, so the spin degeneracy factor $(2S+1)$ is in this case only 2. Using $E = \hbar kc$ we arrive at

$$g(E) dE = \frac{V}{\pi^2 \hbar^3 c^3} E^2 dE, \quad (30.47)$$

and hence

$$U = \int_0^\infty E g(E) dE = \frac{V}{\pi^2 \hbar^3 c^3} \int_0^\infty \frac{E^3 dE}{z^{-1} e^{\beta E} - 1}, \quad (30.48)$$

and using

$$\int_0^\infty \frac{E^3 dE}{z^{-1}e^{\beta E} - 1} = (k_B T)^4 \Gamma(4) \text{Li}_4(z), \quad (30.49)$$

and recognizing that $z = 1$ because $\mu = 0$ and hence $\text{Li}_4(z) = \zeta(4) = \pi^4/90$, and using $\Gamma(4) = 3! = 6$, we have that

$$U = \frac{V\pi^2}{15h^3 c^3} (k_B T)^4, \quad (30.50)$$

which agrees with eqn 23.37.

For Bose systems with a dispersion relation like $E = \hbar^2 k^2 / 2m$ (i.e. for a gapless dispersion, where the lowest-energy level, corresponding to $k = 0$ or infinite wavelength, is at zero energy), the chemical potential has to be negative. If it were not, the level at $E = 0$ would have infinite occupation. Thus $\mu < 0$, and hence the fugacity $z = e^{\beta\mu}$ must lie in the range $0 < z < 1$. But what value will the chemical potential take?

Equation 30.45 can be rearranged to give

$$\frac{n\lambda_{\text{th}}^3}{2S+1} = \text{Li}_{3/2}(z), \quad (30.51)$$

and here we hit an uncomfortable problem. The left-hand side can be increased if $n = N/V$ increases or if T decreases (because $\lambda_{\text{th}} \propto T^{-1/2}$). We can plug numbers for n and T into the left-hand side and then read off a value for z from the graph in Fig. 30.3, which shows the behaviour of the function $\text{Li}_{3/2}(z)$ (and also $\text{Li}_{5/2}(z)$). As we raise n or decrease T , we make the left-hand side of eqn 30.51 bigger and hence z bigger, so that μ becomes less negative, approaching 0 from below. However, if

$$\frac{n\lambda_{\text{th}}^3}{2S+1} > \zeta\left(\frac{3}{2}\right) = 2.612, \quad (30.52)$$

there is no solution to eqn 30.51. What has happened?

30.4 Bose–Einstein condensation (BEC)

The solution to the conundrum raised in the previous section is remarkably subtle, but has far-reaching consequences. As the chemical potential has become closer and closer to zero energy, approaching this from below, the lowest energy level has become macroscopically occupied. The reason our mathematics has broken down is that our usual, normally perfectly reasonable, approximation in going from a sum to an integral in evaluating our grand partition function is no longer valid.

In fact, we can see when this fails using a rearranged version of eqn 30.52. Failure occurs when we fall below a temperature T_c given by

$$k_B T_c = \frac{2\pi\hbar^2}{m} \left(\frac{n}{2.612(2S+1)} \right)^{2/3}. \quad (30.53)$$

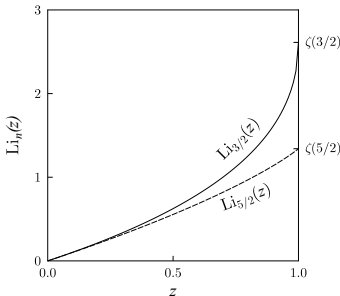


Fig. 30.3 The functions $\text{Li}_{3/2}(z)$ and $\text{Li}_{5/2}(z)$. For $z \ll 1$ (the classical regime), $\text{Li}_n(z) \approx z$. Also, $\text{Li}_n(1) = \zeta(n)$.

We can perform a corrected analysis of the problem as follows. We separate N into two terms:

$$N = N_0 + N_1, \quad (30.54)$$

where N_0 is

$$N_0 = \frac{1}{1 - e^{\beta\mu}} = \frac{z}{1 - z}, \quad (30.55)$$

the number of particles in the ground state, and N_1 is our original integral representing all the other states. Thus above T_c ,

$$N = N_1 = \frac{(2S+1)V}{\lambda_{\text{th}}^3} \text{Li}_{3/2}(z), \quad (30.56)$$

but below T_c , N_1 is fixed to be

$$N_1 = \frac{(2S+1)V}{\lambda_{\text{th}}^3} \text{Li}_{3/2}(1), \quad (30.57)$$

so that the concentration of particles in the excited state is

$$n_1 \equiv \frac{N_1}{V} = \frac{(2S+1)\zeta(\frac{3}{2})}{\lambda_{\text{th}}^3}. \quad (30.58)$$

Any remaining particles must be in the ground state, so that

$$n \equiv \frac{N}{V} = \frac{(2S+1)\zeta(\frac{3}{2})}{\lambda_{\text{th}}(T_c)^3}. \quad (30.59)$$

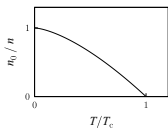


Fig. 30.4 The number of particles in the ground state as a function of temperature, after eqn 30.60.

⁸This is often abbreviated to BEC.

Hence

$$\frac{n_0}{n} = \frac{n - n_1}{n} = 1 - \left(\frac{T}{T_c}\right)^{3/2}. \quad (30.60)$$

This function is plotted in Fig. 30.4 and shows how the number of particles in the ground state grows as the temperature is cooled below T_c . This macroscopic occupation of the ground state is known as **Bose–Einstein condensation**.⁸ Note that this transition is not driven by interactions between particles (as we had for the liquid–gas transition); we have so far only considered non-interacting particles; the transition is driven purely by the requirements of exchange symmetry on the quantum statistics of the bosons.

The term ‘condensation’ often implies a condensation in space, as when liquid water condenses on a cold window in a steamy bathroom. However, for Bose–Einstein condensation it is a condensation in k -space, with a macroscopic occupation of the lowest energy state occurring below T_c .

Example 30.8

Find the internal energy $U(T)$ at temperature T for the Bose gas.

Solution:

The internal energy of the system only depends on the excited states, since the macroscopically occupied ground state has zero energy. Since $z = 1$ for $T \leq T_c$, we have that

$$\begin{aligned} U &= \frac{3}{2} N_1 k_B T \frac{\zeta(\frac{5}{2})}{\zeta(\frac{3}{2})} \\ &= \frac{3}{2} N k_B T \frac{\zeta(\frac{5}{2})}{\zeta(\frac{3}{2})} \left(\frac{T}{T_c}\right)^{3/2} \\ &= 0.77 N k_B T_c \left(\frac{T}{T_c}\right)^{5/2}. \end{aligned} \quad (30.61)$$

For $T > T_c$ we have (from eqn 30.46)

$$U = \frac{3}{2} N k_B T \frac{\text{Li}_{5/2}(z)}{\text{Li}_{3/2}(z)}. \quad (30.62)$$

This example gives the high-temperature results as a function of the fugacity, but z is temperature-dependent. For a system with a fixed number N of bosons, we can extract z via $N/V = (2S + 1)\text{Li}_{3/2}(z)/\lambda_{\text{th}}^3$ and equating this with eqn 30.59 yields

$$\frac{T}{T_c} = \left[\frac{\zeta(\frac{3}{2})}{\text{Li}_{3/2}(z)} \right], \quad (30.63)$$

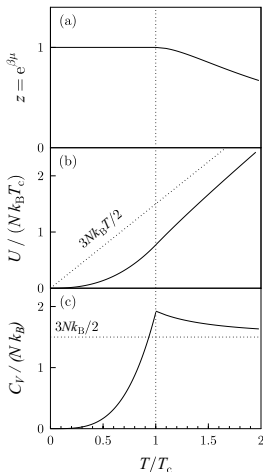


Fig. 30.5 The (a) fugacity, (b) internal energy and (c) heat capacity for a system of bosons as a function of temperature.

which although it cannot be straightforwardly inverted to make z the subject, does show how z is related to T above T_c . (Below T_c , z is practically one.)

The fugacity z , internal energy U and heat capacity C_V , calculated for non-interacting bosons, are plotted in Fig. 30.5. The fugacity is obtained by numerical inversion of eqn 30.63; it rises up towards unity as you cool, and below T_c is not actually one but very close to it. The internal energy U in Fig. 30.5(a) is obtained from eqn 30.62, while the heat capacity C_V is plotted from eqn 30.66, to be proven in the exercises at the end of this chapter.

The Indian physicist S. N. Bose wrote to Einstein in 1924 describing his work on the statistical mechanics of photons. Einstein appreciated the significance of this work and used Bose's approach to predict what

is now called Bose–Einstein condensation.

In late 1930's, it was discovered that liquid ^4He becomes a **superfluid** when cooled below about 2.2 K. Superfluidity is a quantum-mechanical state of matter with very unusual properties, such as the ability to flow through very small capillaries with no measurable viscosity. Speculation arose as to whether this state of matter was connected with Bose–Einstein condensation.

Example 30.9

Estimate the Bose–Einstein condensation temperature for liquid ^4He , given that $m_{\text{He}} \approx 4m_p$ and that the density $\rho \approx 145 \text{ kg m}^{-3}$.

Solution:

Using $n = \rho/m$, eqn 30.53 yields $T_c \approx 3.1 \text{ K}$ which is remarkably close to the experimental value of the superfluid transition temperature.

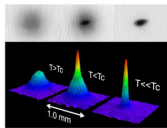


Fig. 30.6 Observation of Bose–Einstein condensation by absorption imaging. The data are shown as shadow pictures (upper panel) and as a three-dimensional plot (lower panel); the blackness of the shadow in the upper panel is here represented by height in the lower panel. These pictures measure the slow expansion of the trapped atoms observed after a 0.006 s time of flight, and thus measure the momentum distribution inside the cloud. The left-hand picture shows an expanding cloud cooled to just above the transition point. In the right-hand picture we see the velocity distribution well below T_c where almost all the atoms are condensed into the zero-velocity peak. (Image courtesy W. Ketterle.)

⁹Alkali atoms are in Group I of the periodic table and include Li, Na, K, Rb and Cs.

¹⁰The 2001 Nobel Prize was awarded to Eric Cornell and Carl Wieman (who did the experiment with rubidium atoms) and to Wolfgang Ketterle (who did it with sodium atoms).

Despite the agreement between this estimate and the experimental value, things are a bit more complicated. The particle density of ^4He is very high and interactions between helium atoms cannot be ignored; ^4He is a strongly interacting Bose gas, and therefore the predictions of the theory outlined in this chapter have to be modified.

A more suitable example of Bose–Einstein condensation is provided by the very dilute gases of alkali metal atoms⁹ that can be prepared inside magnetic ion traps. The atoms, usually about 10^4 – 10^6 of them, can be trapped and cooled using the newly developed techniques of laser cooling. These alkali atoms have a single electronic spin due to their one valence electron and this can couple with the non-zero nuclear spin. Each atom therefore has a magnetic moment and thus can be trapped inside local minima of magnetic field. The density of these **ultracold atomic gases** inside the traps are very low, more than seven orders of magnitude lower than that in ^4He , though their masses are higher. The Bose–Einstein condensation temperature is therefore also very low, typically 10^{-8} – 10^{-6} K , but these temperatures can be reached using laser cooling. The low density precludes significant three-body collisions (in which two atoms bind with the third taking away the excess kinetic energy, thus causing clustering), but two-body collisions do occur which allow the cloud of atoms to thermalize. Example data are shown in Fig. 30.6 from one such experiment which clearly show that below a critical temperature Bose–Einstein condensation is taking place.¹⁰

Superfluidity is also found in these ultracold atomic gases; it turns out that the very weak interactions that exist between the alkali atoms are important for this to occur (a non-interacting Bose gas does not show superfluidity). Other experiments have explored the intriguing consequences of **macroscopic quantum coherence**, the property that in the condensed state all the atoms exist in a coherent quantum superposition.

Electrons do not exhibit Bose–Einstein condensation because they are fermions, not bosons, but they can show other condensation effects such as **superconductivity**. In a superconductor, a weak attractive interaction (which can be mediated by phonons) allows pairs of electrons to form **Cooper pairs**. A Cooper pair is a boson, and the Cooper pairs themselves can form a coherent state below the superconducting transition temperature. Many common superconductors can be described in this way using the **BCS theory of superconductivity**,¹¹ though many newly discovered superconductors, such as the **high-temperature superconductors** which are ceramics, do not seem to be described by this model.

¹¹BCS is named after its discoverers, John Bardeen, Leon Cooper, and Robert Schrieffer.

Chapter summary

- Non-interacting bosons can be described using the equations

$$\begin{aligned} N &= \frac{(2S+1)V}{\lambda_{\text{th}}^3} [\mp \text{Li}_{3/2}(\mp z)], \\ U &= \frac{3}{2} N k_B T \frac{\text{Li}_{5/2}(\mp z)}{\text{Li}_{3/2}(\mp z)}, \\ \Phi_G &= -\frac{2}{3} U. \end{aligned}$$

- In a Fermi gas (a gas of fermions), electrons fill states up to E_F at absolute zero. At non-zero temperature, electrons with $k_B T$ of E_F are important in determining the properties.
- The results for a Fermi gas can be applied to the electrons in a metal.
- In a Bose gas, Bose–Einstein condensation can occur below a temperature given by

$$k_B T_c = \frac{2\pi\hbar^2}{m} \left(\frac{n}{2.612(2S+1)} \right)^{2/3}.$$

- The results for a Bose gas can be applied to liquid ^4He and dilute ultracold atomic gases.

Further reading

For further information, see Ashcroft and Mermin (1976), Annett (2004), Foot (2004), Ketterle (2002) and Pethick and Smith (2002).

Exercises

- (30.1) Show that in the classical limit, when the fugacity $z = e^{\beta\mu} \ll 1$, z is the ratio of the thermal volume to the volume per particle of a single-spin excitation.

- (30.2) Show that the pressure p exerted by a Fermi gas at absolute zero is

$$p = \frac{2}{5} n E_F, \quad (30.64)$$

where n is the number density of particles.

- (30.3) Show that for a gas of fermions with density of states $g(E)$, the chemical potential is given by

$$\mu(T) = E_F - \frac{\pi^2}{6} (k_B T)^2 \frac{g'(E_F)}{g(E_F)} + \dots \quad (30.65)$$

- (30.4) Show that the heat capacity of a system of non-interacting bosons is given by

$$\begin{aligned} C_V &= \frac{15}{4} \frac{\zeta(\frac{5}{2})}{\zeta(\frac{3}{2})} N k_B \left(\frac{T}{T_c} \right)^{3/2}, & T < T_c, \\ C_V &= \frac{3}{2} N k_B \left(\frac{5}{2} \frac{\text{Li}_{5/2}(z)}{\text{Li}_{3/2}(z)} - \frac{3}{2} \frac{\text{Li}_{3/2}(z)}{\text{Li}_{1/2}(z)} \right), & T > T_c. \end{aligned} \quad (30.66)$$

- (30.5) Show that Bose-Einstein condensation does not occur in two dimensions.

- (30.6) In Bose-Einstein condensation, the ground state becomes macroscopically occupied. What about the first excited state, which might be only a small energy above the ground state; is it also macroscopically occupied?